CREATING KPI CARDS, FORMATTING & DAX-BASED MEASURES IN POWER BI

1. Introduction

In any business analytics report, Key Performance Indicators (KPIs) play a central role in summarizing the most important metrics in a compact, easy-to-understand manner. In Power BI, one of the simplest and most commonly used visuals for displaying KPIs is the **Card visual**, which shows a single aggregated number such as total orders, total sales, total profit, average discount, or profitability.

The process of creating KPI cards involves:

- Selecting or creating the right measure,
- Adding a card visual,
- Applying consistent formatting,
- Setting display units,
- Applying currency or percentage formatting, and
- Organizing the visuals effectively on the dashboard.

YATICORP
EMPOWERING EDUCATION

In addition, Power BI provides capabilities to create new calculated values using **DAX (Data Analysis Expressions)** when the desired KPI is not directly available from existing fields.

The following notes deeply explain how KPI cards are created, formatted, duplicated, and enhanced using DAX for advanced metrics such as profitability.

---

2. Creating the First KPI – Total Orders

The dataset in the example consists of tables containing order details from different years (2018–19 and 2020–21). After preparing the model and appending the tables into a single combined dataset called *Append1*, the first KPI to be created is **Total Orders**.

Steps to Create Total Orders KPI:



1. **Add a Card Visual**
    - From the "Visualizations" pane, select the **Card** visual.
    - Drag it onto the report canvas.
2. **Assign the Appropriate Field**
    - Under the *Fields* pane, locate the column **Order ID**.
    - Use **Count** or **Count Distinct** as applicable.
    - Typically, *Count Distinct* is used when each order ID must be counted uniquely.
3. **Formatting the Card Visual**
    - Go to **Format visual → General → Title**.
    - Turn *Title* ON.
    - Enter the title as **"Total Orders"**.
    - Adjust text size, font, color, and alignment if required.
4. **Placement & Optimization**
    - Adjust the card's size and place it appropriately within the KPI section of the dashboard.

This completes the first KPI.

YATICORP
EMPOWERING EDUCATION

3. Creating Total Sales Using Card Duplication (Efficient Method)

Instead of creating a new card from scratch, Power BI allows copying and pasting visual elements using **Ctrl+C and Ctrl+V**. This helps maintain consistent formatting and reduces repetitive formatting work.

Steps:

1. Select the card visual for Total Orders.
2. Press **Ctrl+C** and then **Ctrl+V** to make a duplicate.
3. Move the duplicated card next to the original.
4. Rename the **Title** to **Total Sales**.
5. Remove the existing field from the card:
    - In the *Build visual* section, uncheck **Order ID**.
6. Add the **Sales** measure/column:
    - Check the box next to **Sales**.

YATICORP
EMPOWERING EDUCATION

Handling Currency Format

Sometimes the Sales field displays values as Indian Rupees (₹) or without proper formatting. To correct this:

1. Click on the **Sales** field in the Fields pane.
2. In the ribbon, choose **Currency format**.
3. Select **English (United States)** to display values in **USD ($)**.
4. Choose appropriate display units:
    - Auto
    - Thousands
    - Millions
    - None (exact value)

The Sales KPI is now properly displayed.

4. Creating the Total Profit KPI

This follows the same duplication method used for Total Sales:

Steps:

1. Copy and paste an existing KPI card.
2. Change the title to **Total Profit**.
3. Remove the previously used field (e.g., Sales).
4. Check the **Profit** field.
5. Adjust the number formatting and card placement.

This provides the Total Profit KPI.

---

5. Creating the Total Quantity KPI

Once again using duplication for efficient formatting consistency:

YATICORP
EMPOWERING EDUCATION

Steps:

1. Duplicate an existing card.
2. Change the title to **Total Quantity**.
3. Remove previously assigned field(s).
4. Check the **Quantity** field.
5. Adjust the **Display Units** under *Callout Value*:
    - If Auto displays values in thousands but full value is required,
    - Change Display units → **None**.

The Total Quantity KPI is now ready.

---

6. Creating the Average Discount KPI

This KPI is slightly different because the raw discount field usually contains individual discount percentages per order line. Adding this field to a card visual **by default shows the SUM**, not the average.

Steps to Compute Average Discount Using Card Options:

1. Duplicate an existing card.
2. Change the title to **Average Discount**.
3. Remove the existing field from the card.
4. Check **Discount** from the Fields pane.
5. Power BI will display **Sum of Discount** by default.
6. To change this:
    - Click on the drop-down arrow next to “Sum of Discount”.
    - Choose **Average**.
7. To show the value as a **percentage**:
    - Select the **Discount** field.
    - On the ribbon, click the **Percentage (%)** format option.

YATICORP
EMPOWERING EDUCATION

Once done, the KPI card will display the average discount (e.g., **15.56%**).

Important Insight

- Excel requires manually computing average = total/number of entries.
- Power BI allows selecting **Average** directly due to built-in aggregation support.

---

7. Creating the Profitability KPI (Using DAX Measure)

Unlike previous KPIs that directly use aggregated columns, *Profitability* requires computing:

Profitability=∑(Profit)∑(Sales)\text{Profitability} = \frac{\sum(\text{Profit})}{\sum(\text{Sales})}Profitability=∑(Sales)∑(Profit)

This is a **ratio** and requires a **DAX measure**.

7.1 Understanding the Need for a DAX Measure

Power BI aggregates columns automatically, but for:

- Ratios,
- Percentages,
- Custom calculated KPIs,
- Logical metrics based on business rules,

we must write a DAX formula.

Profitability is not present as a column in the dataset. It must be computed manually.

YATICORP
EMPOWERING EDUCATION

7.2 Exploring the Data View

Before writing the measure:

- Open the **Data View** (second icon on left panel).
- Inspect the **Append1** table to verify:
    - Profit column
    - Sales column
    - Order IDs (possibly repeated due to multiple products per order)
- Understand that the total profit and total sales must be aggregated using **SUM()**.

7.3 Creating a New Measure Using DAX

Steps:

1. Select the table (e.g., *Append1*) in the Fields pane.
2. A new tab **Table Tools** appears in the ribbon.
3. Click **New Measure**.
4. In the DAX formula bar, type:

```
Profitability =
   SUM(Append1[Profit]) / SUM(Append1[Sales])
```

Explanation of DAX Code:

- SUM(Append1[Profit])
  Aggregates all profit values across the dataset.
- SUM(Append1[Sales])
  Aggregates all sales values across the dataset.
- Dividing the two gives the overall profit margin.

YATICORP
EMPOWERING EDUCATION

7.4 Formatting the Measure

After pressing Enter:

- The new measure appears under the Append1 table.
- Select the measure.
- In the ribbon, click the **Percentage** (%) formatting option.

Power BI will automatically multiply the value by 100 when displayed.

For example:

- If the ratio = 0.1274
- Displayed result = **12.74%**

7.5 Displaying the Profitability KPI Using a Card

Steps:

1. Duplicate an existing KPI card.
2. Change the title to **Profitability**.
3. Remove the existing field.
4. Check the **Profitability** measure.
5. Adjust positioning on the canvas.

This completes the Profitability KPI creation.

---

8. Summary of KPI Creation Workflow

KPI	Approach	Field/Measure Used	Notes
Total Orders	Card	Count Distinct of Order ID	First KPI
Total Sales	Card	Sum of Sales	Currency formatting needed
Total Profit	Card	Sum of Profit	Similar to Total Sales
Total Quantity	Card	Sum of Quantity	Display units adjustment
Average Discount	Card	Average of Discount	Change aggregation + %
Profitability	Card + DAX	SUM(Profit)/SUM(Sales)	Must create new measure

---

YATICORP
EMPOWERING EDUCATION

9. Understanding Aggregations in Power BI

Power BI provides multiple aggregation functions including:

- Sum
- Average
- Minimum
- Maximum
- Count
- Distinct Count
- Standard Deviation
- Variance
- Median

YATICORP
EMPOWERING EDUCATION

Choosing the correct aggregation is critical for accurate KPI results. For example:

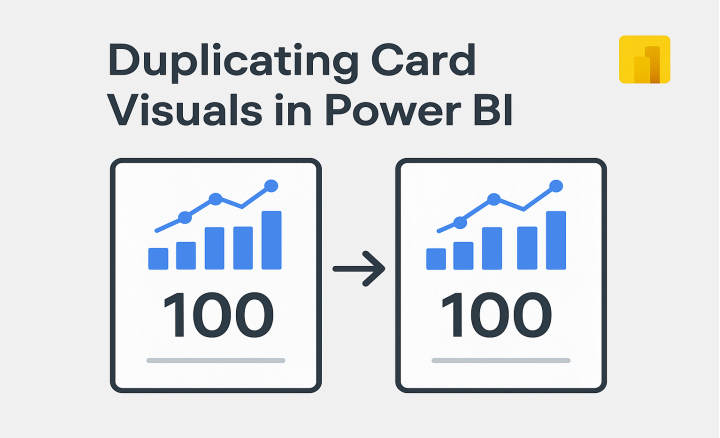
- Discount must be averaged, not summed.
- Sales and Profit are always summed.
- Order IDs often require distinct count.

---

10. Advantages of Duplicating Card Visuals

Duplicating visuals is a powerful technique because it:

- Saves formatting time,
- Maintains consistency across all KPIs,
- Provides a clean professional layout,
- Reduces repetitive manual work.



This is a best practice for dashboard building.

YATICORP
EMPOWERING EDUCATION

11. Styling and Formatting Considerations

For well-presented KPIs:

- Use consistent **font sizes**,
- Keep **number formats uniform** (thousands vs none),
- Apply **title casing** (Total Orders, Total Sales, etc.),
- Align visuals in a straight horizontal or grid layout,
- Hide unnecessary panes (e.g., Filters) to enhance visual clarity.

---

12. Importance of Measures vs Calculated Columns

- **Calculated columns** are computed row by row and stored in the model.
- **Measures** are computed dynamically based on filter context.

Profitability is a measure because it:

- Depends on aggregated totals,
- Changes dynamically based on slicer selections,
- Should not be calculated row by row.

Measures are more efficient and correct for KPI calculations.

---

13. Final Thoughts

YATICORP
EMPOWERING EDUCATION

The process described demonstrates key Power BI skills:

- Using card visuals efficiently,
- Handling aggregations,
- Applying number and currency formats,
- Understanding data structures,
- Creating DAX measures for advanced KPIs.

This workflow is commonly used in professional dashboards across industries such as sales, logistics, supply chain, finance, and operations.

By combining visual duplication, proper aggregation selection, and DAX, analysts can quickly produce powerful, accurate, and visually consistent KPI dashboards.